Near-side azimuthal and …

I. INTRODUCTION

33-35. Systematic studies … revealed significant modification.”

35-38. The associated particle distribution … is also significantly modified …”

Comment regarding both of the above: When something is “modified” it is meaningful only when the one knows to what the present “modified” quantity/distribution is being compared. Maybe that is “common knowledge” to those working in the field, but it is not obvious to me as a non-expert and I should think that it would not be difficult to make this more obvious to the non-expert.

39-41. “… there is a peak that is narrow in azimuth (delta-phi) and in pseudorapidity … refer to as the “jet-like correlation”.

48-51. “… at intermediate transverse momentum ~~range~~, the mechanism for producing hadrons seems to differ …”

50-52. “In central Au+Au collisions … relative to what is observed in p+p …”

52-54. “…measured baryon to meson production ratios in Au+Au collisions increase with pT …” (pT of what?) “… until reaching ~~their~~ a maximum value …”

55-58. “A fall-off …meson production ratio is observed … non-strange baryon to meson production ratios in Au+Au … approach the corresponding values …”

62. “… GeV as measured by the STAR … “

67. “…unidentified charged hadron two-particle correlations in the same collision systems.” [Or, I have misunderstood what you intended to say.]

67-70. “~~Both identified~~ Events that triggered on identified neutral strange particles, associated with unidentified charged hadrons (K0s-h…), and, events that triggered on unidentified charged hadrons, associated with identified neutral strange particles (h-K0s, …) were both studied.

77. “Results are …” => “The results are …”

II. EXPERIMENTAL SETUP AND PARTICLE RECONSTRUCTION

80-83. "… are based on the analysis … respectively." New Paragraph

111-125. I suggest that these lines be moved to the New Paragraph following "respectively." The logic to have these here seems much more reasonable than their present location.

New Paragraph (The numbering will change after the 15 lines are moved.)

83. "For d+Au collisions …"

88. "… Au beam direction ~~and~~ thereby accepting …"

93-97. "The MB Au+Au events required a coincidence between the two ZDCs, a signal in both BBCs and … scintillator slats aligned parallel to the beam axis and arranged … interactions."

97-102. "An additional online trigger was used for central Au+Au collisions so as to sample the most central 12% of the total hadroncic cross section. This trigger was based on … in the CTB. ~~The central trigger … cross section~~."

128. I suggest replacing "daughters" with "decay products"

134-135. "… for each data set ~~and~~ so as to yield a signal to background ratio of at least 15:1."

135-140. "~~For the analysis presented here … simply as (lambda) baryons~~." Replace with the following: "For the analyses presented here the results obtained with the (lambda) or the (lambda-bar) trigger, each with associated charged hadrons, showed no significant differences. Therefore, to increase the statistical significance of our results, the correlations for the (lambda) and the (lambda-bar) were added together and, for the remainder of this discussion are referred to as (lambda) baryon results."

III. METHOD

A. Correlation technique

145-146. "… that trigger particle in pseudorapitidy …"

B. Single charged track efficiency correction

160. "For unidentified associated charged particles, …"

171. "For identified associated strange particles, …." (Note the word order.)

182-185. "The systematic uncertainty ~~on~~ associated with the efficiency … hadrons is 5% and is strongly … data sets."

185. "For identified associated particle ratios, [add comma] …"

190. "… to secondary Λ particles from …"

C. Pair acceptance correction

200. "The requirement ~~of~~ that each measured track falls within …"

202. "The track pair geometric acceptance depends on pseudorapidity and is observed to be $\approx$ 100% for $\Delta\eta \approx 0$, ~~however~~ whereas, near …"

204. "The track pair azimuthal acceptance is …"

D. Track merging correction

219-220. "… dip at small $\Delta\phi$ and small $\Delta\eta$ due to …"

226. "… ~~closer~~ more nearly the same …"

231. "In ~~a~~ [remove "a"] lambda decay, the proton ~~daughter~~ carries more of the Λ momentum than the pion ~~daughter~~. Therefore …" ["daughter" is unnecessary.]

233-234. "… particles with lower ~~momenta~~ momentum."

234-235. "~~Since~~ Because track merging … particles, and, because the signal sits …"

237. "~~Since the dip~~ …" "Moreover, because the dip …"

243-244. "… and using the fit to extract the yield." => "and the fitted function is used to extract the yield."

E. Yield extraction

255-256. "~~Extraction of the yield in this paper follows the method and the notation in [7,12]."~~ => "The method and the notation employed to extract the yield for this analysis is described in references 7 and 12."

264-265. "… consistent with references 12 and 30-32."

266+ (numbering is absent for awhile) "This means that the jet-like yield determined by … equation … were b… is a constant offset ~~determined~~ obtained by fitting … ($\Delta\eta$).” [What I don’t like about this sentence is that it does not say to what the Gaussian + background term is fitted. I think this is essential.]

267+ “~~Sample correlations are given in Figure 1~~.” “Illustrative corrected correlations are shown in Figure 1.”

267+ “… using bin counting …” [Just what is “bin counting”?] Presumably you mean “… we integrate Equation 4 over $\Delta\eta$ using the bin contents to determine …”

268-271. “When it is not possible, the yield from the Gaussian fit is used …” [To my reading this sounds very redundant with what is said just below 267. Is this really needed? What does it add to what has already been said?]

IV. RESULTS

A. Charged V^0 correlations

274. “Previous studies have demonstrated …”

277. “~~This indicates~~ that …” => “These observations imply that …”

281. “… production of particles.” [New paragraph here.]

283. “The statistics of the …” => “In the measurements reported here, the statistics of the …” [The new paragraph and these words make it clear that you have transitioned from “Previous studies” to the results reported in this paper.]

286. “was ~~only~~ possible to determine the composition of the jet-like correlation only in Cu+Cu collisions.”

288. “These ~~data~~ measurements are shown in Figure 2 and compared to results from inclusive …” [The term “data” is so innocuous it is unhelpful. What you are showing are your measurements, i.e., the results of your analysis to get these measurements.]

290. “… and from Alice [35]. Also shown are the results from PYTHIA [36] ~~from~~ using the Perugia … tunes.” [Break the long and confusing sentence as shown.]

291-292. “The Cu+Cu ~~data~~ results are consistent … from *p+p* collisions.”

293-297. You observe that the Cu+Cu (jet-like) results are consistent with the (strange particle production) results from p+p. Then you say that “This [whatever “this” refers to ??] further supports what you have already said (earlier) that the jet-like correlation is dominantly produced by fragmentation.” The rest of the logic I find very confusing and circular. Perhaps you can explain it to me in a way that I can appreciate what "This" is and how "this" supports the claim that the jet-like correlation is dominantly produced by fragmentation.

297. "This indicates that …" What, again, is "this"?

302-303. "As seen in Fig. 2, this … ratio, measured using jet-like correlations at STAR, is also consistent ~~between~~ with inclusive measurements at different energies." [Well, I think that this is what you are saying, but correct me if it is wrong.]

305-308. "While PYTHIA is able to describe the π and ω production [38,39], it generally underestimates the production of strange particles and baryons, specifically the K^0_S and the Λ [34, … 39]."

309-312. "As seen in Figure 2, PYTHIA underestimates … ratio for Tune A and the Perguia tunes." [I do not understand to what the next phrase applies: "which improved the description of inclusive strangeness production at LHC energies." Please explain.]

313. "… at higher energies…" [This is a vague, not-specific description and, as used, unhelpful. "higher energies" begs the clarification "higher than what?" Please explain.]

314. "For each of the PYTHIA tunes used, the predicted value of the [insert ratio here] ratio obtained for inclusive production and for the jet-like correlation mechanism is lower than the corresponding measured ratios. "

319-321. [The next sentence I find very confusing. The first phrase is merely a restatement of the preceding sentence and I do not see how that observation "**predicts**" that both the inclusive and the jet-like correlation are dominated by fragmentation. Perhaps it is my lack of depth here, so if you can explain this to me it would be helpful.]

322. "The h-K^0_S and …" [It seems odd to begin a sentence with this notation.]

323. "… light quark jets [???]," [I think that you should have a reference here.]

323-330. [I am afraid that I am completely unable to judge whether this section is either valid or appropriate here under "RESULTS". Is this speculation or established physics? For example, "It may be easier for a PHYTHIA to describe …" sounds more like something that would come in a section (not now in the paper) called "DISCUSSION".] In any case, remove the "a" before PYTHIA on line 327.

Table II. (a) the line spacing seems odd or misadjusted. (b) I suggest that for consistency and ease of reading you put the "0-60%" over the Cu-Cu columns and

"Min bias" – or something equivalent – over the d+Au columns. (c) There appear to be "-" entries where there is no measured yield but not in all places. Why?

Fig. 2: Why is there no mention of the PYTHIA curves in the caption? You might include something like "Results from PYTHIA calculations are shown with different tunes as described in the text."

Fig. 3: (a) The column headings for the symbols are a bit odd. Could you not put the centrality below each heading for Cu-Cu and Au-Au so as to be consistent and more accommodating for the reader? (b) I prefer "measurements" instead of "data": "The measurements are compared …" and "The Measured values have been offset …"

336. "The ~~data~~ measurements are tabulated …"

351. "~~Data~~ The measurements are comparable …"

353-356. "We focus on the Perguia 2011 tune because this is the most recent tune that incorporated results on strangeness production at the LHC." [What is the merit of the next phrase? Is it important and essential or can it be dropped? It appears to be an "aside" or "parenthetic comment". Moreover, it does not seem to fit as a reason why you focus on the Perguia 2011 tune so it does not fit in that sentence. It may just be a parenthetic comment and could be included as such explicitly.]

Fig. 4. (a) "The ~~data~~ measurements are compared …" (b) There is a double period (one black, one red) following GeV.

362. "Next the jet-like yields are studied as a function …" => "The jet-like yields were also studied as a function …"

368-369. "… bin counting …" How about this: "All yields were determined by counting events in each N_{part} bin."

Fig. 4. You make not mention of the curves in the caption. How about adding "Results from PYTHIA calculations are shown as described in the text."

372. "These ~~data~~ results are compared …"

376. "All yields were determined by counting events in each $p_T^{associated}$ bin."

380. "These ~~data~~ results are compared …"

Fig. 5. "The ~~data~~ measurements are compared to …"

388. "… ~~however~~ whereas, the contribution …"

389. "… to be small ~~from~~ based on the results …"

390. "The jet-like yield shows a systematic dependence on particle type with the jet-like yield highest …"

405. "… trigger ~~were~~ was wider than …"

406. "… trigger because the jet-like yield …"

408. "The difference in the particle type ordering could …"

420. "… in the particle type ordering."

422. "… widths of the jet-like yield. This would, however, be an area of interest …"

433. "While the jet-like yield … [7], ~~however~~, the jet-like yield …"

461. "These ~~data~~ results indicate …"

463. "… triggers, ~~however~~, but these data do not …"

V. CONCLUSIONS

No problems.